



Introduction to Bioengineering Theory (BBL1020; 3-0-0)

Lecture-2: 3 classes

*Please listen carefully
&*

Keep your phone(s) on silent mode

Exploring levels of Biological Organization



◀ 1 The Biosphere

Even from space, we can see signs of Earth's life—in the green mosaic of the forests, for example. We can also see the entire **biosphere**, which consists of all life on Earth and all the places where life exists: most regions of land, most bodies of water, the atmosphere to an altitude of several kilometers, and even sediments far below the ocean floor.



◀ 2 Ecosystems

Our first scale change brings us to a North American mountain meadow, which is an example of an ecosystem, as are tropical forests, grasslands, deserts, and coral reefs. An **ecosystem** consists of all the living things in a particular area, along with all the nonliving components of the environment with which life interacts, such as soil, water, atmospheric gases, and light.



▶ 3 Communities

The array of organisms inhabiting a particular ecosystem is called a biological **community**. The community in our meadow ecosystem includes many kinds of plants, various animals, mushrooms and other fungi, and enormous numbers of diverse microorganisms, such as bacteria, that are too small to see without a microscope. Each of these forms of life belongs to a *species*—a group whose members can only reproduce with other members of the group.



▶ 4 Populations

A **population** consists of all the individuals of a species living within the bounds of a specified area. For example, our meadow includes a population of lupine (some of which are shown here) and a population of mule deer. A community is therefore the set of populations that inhabit a particular area.



▲ 5 Organisms

Individual living things are called **organisms**. Each plant in the meadow is an organism and so is each animal, fungus, and bacterium.



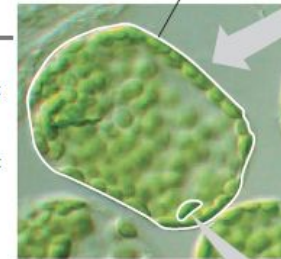
▼ 6 Organs

The structural hierarchy of life continues to unfold as we explore the architecture of a complex organism. A leaf is an example of an **organ**, a body part that is made up of multiple tissues and has specific functions in the body. Leaves, stems, and roots are the major organs of plants. Within an organ, each tissue has a distinct arrangement and contributes particular properties to organ function.



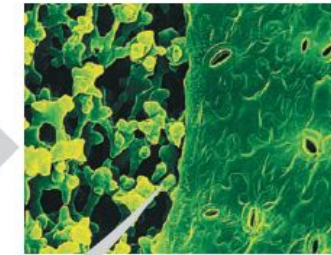
▶ 8 Cells

The **cell** is life's fundamental unit of structure and function. Some organisms consist of a single cell, which performs all the functions of life. Other organisms are multicellular and feature a division of labor among specialized cells. Here we see a magnified view of a cell in a leaf tissue. This cell is about 40 micrometers (μm) across—about 500 of them would reach across a small coin. Within these tiny cells are even smaller green structures called chloroplasts, which are responsible for photosynthesis.



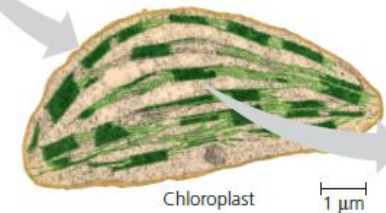
▼ 7 Tissues

Viewing the tissues of a leaf requires a microscope. Each **tissue** is a group of cells that work together, performing a specialized function. The leaf shown here has been cut on an angle. The honeycombed tissue in the interior of the leaf (left side of photo) is the main location of photosynthesis, the process that converts light energy to the chemical energy of sugar. The jigsaw puzzle-like "skin" on the surface of the leaf is a tissue called epidermis (right side of photo). The pores through the epidermis allow entry of the gas CO₂, a raw material for sugar production.



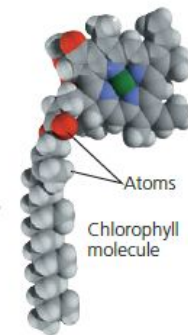
▼ 9 Organelles

Chloroplasts are examples of **organelles**, the various functional components present in cells. The image below, taken by a powerful microscope, shows a single chloroplast.



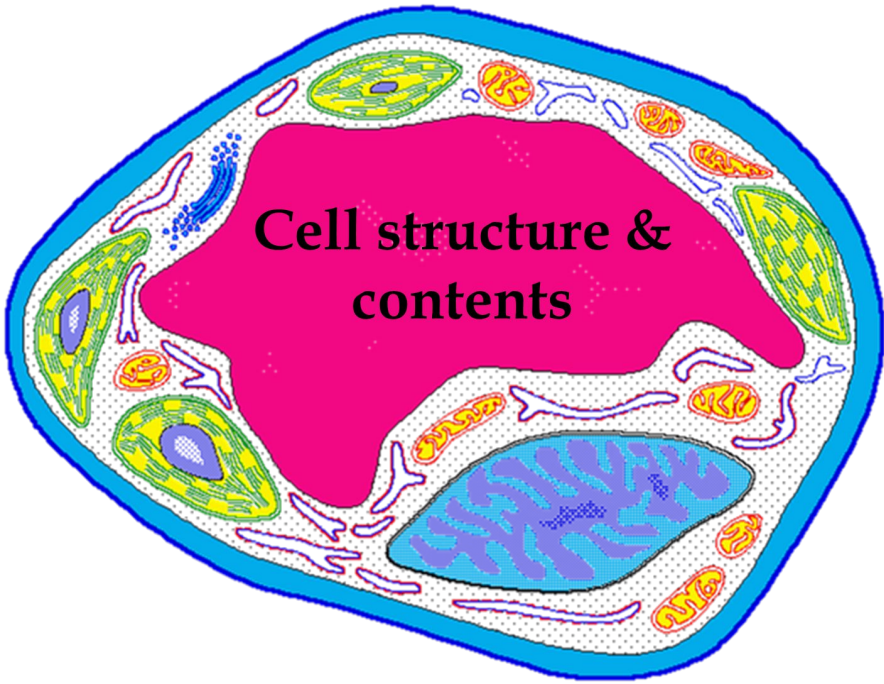
▼ 10 Molecules

Our last scale change drops us into a chloroplast for a view of life at the molecular level. A **molecule** is a chemical structure consisting of two or more units called atoms, represented as balls in this computer graphic of a chlorophyll molecule.

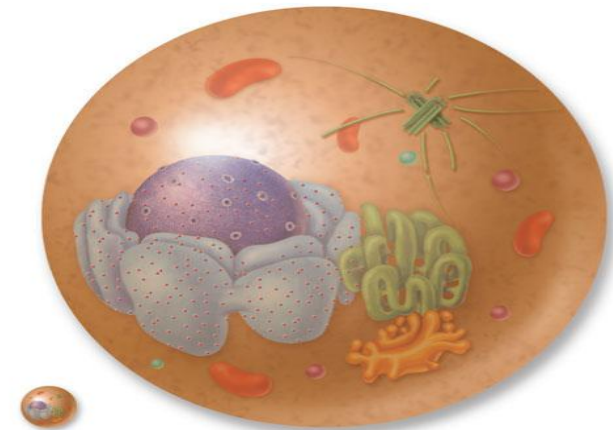


Chlorophyll is the pigment that makes a leaf green, and it absorbs sunlight during photosynthesis. Within each chloroplast, millions of chlorophyll molecules are organized into systems that convert light energy to the chemical energy of food.

Cell Theory



- Cells are the smallest living things. All organisms are composed of cells. They arise only from pre-existing cells.
- Cell size is limited – as the cell size increases, it takes longer for a material to diffuse from the cell membrane to the interior of the cell. As the cell size is increased, the volume increases 10 times faster than the surface area. Light and electron microscopes can aid visualization of cells.

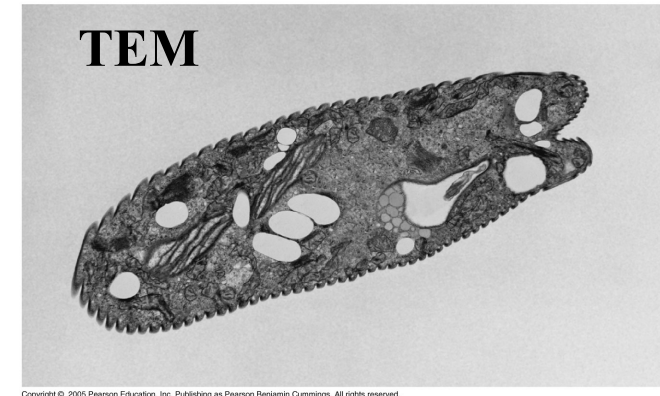
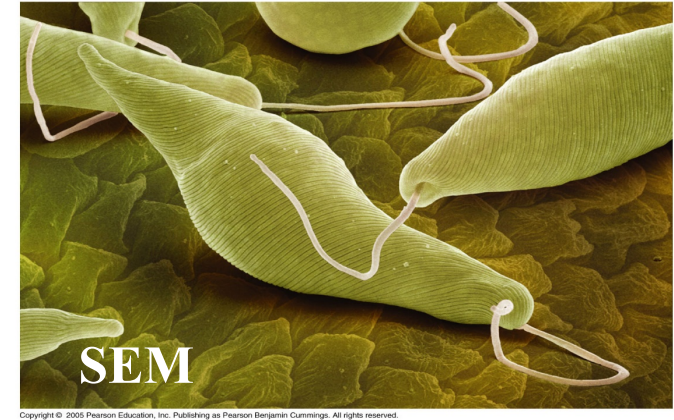


Cell radius (r)	1 unit	10 unit
Surface area ($4\pi r^2$)	12.57 unit ²	1257 unit ²
Volume ($\frac{4}{3}\pi r^3$)	4.189 unit ³	4189 unit ³
Surface Area / Volume	3	0.3

Cell Theory

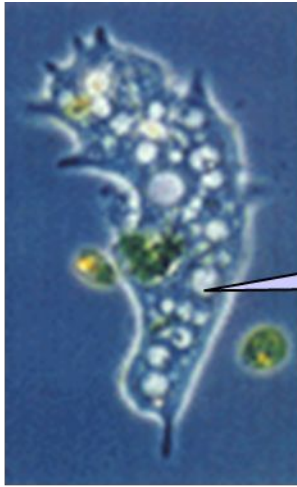


- All cells have genetic material – in a nucleoid or nucleus, cytoplasm – a semifluid matrix, and plasma membrane - phospholipid bilayer.
- A cell is the smallest unit that can perform life functions.
- Light microscope - Can observe living cells in true color; Magnification of up to ~1000x; Resolution ~ 0.2 microns – 0.5 microns.
- Electron microscopes - Images are black and white – may be colorized; Magnification up to ~100,000; Transmission electron microscope (TEM) - 2-D image; Scanning electron microscope (SEM) - 3-D image.

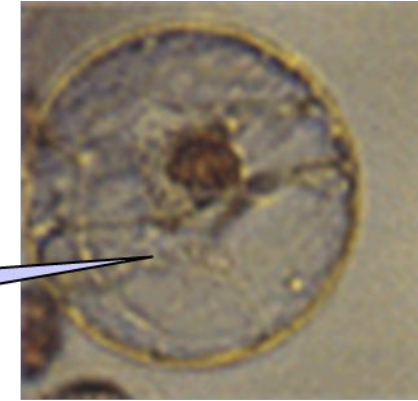


Examples of Cells

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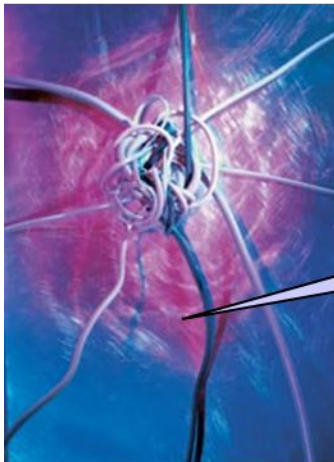
Amoeba Proteus



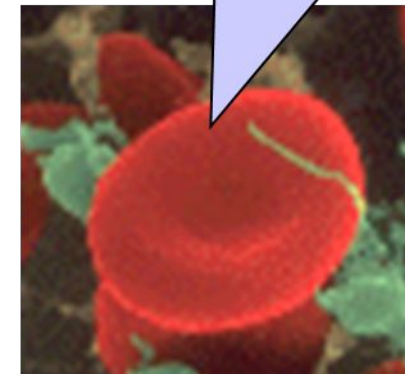
Plant Stem



Bacteria



Nerve Cell

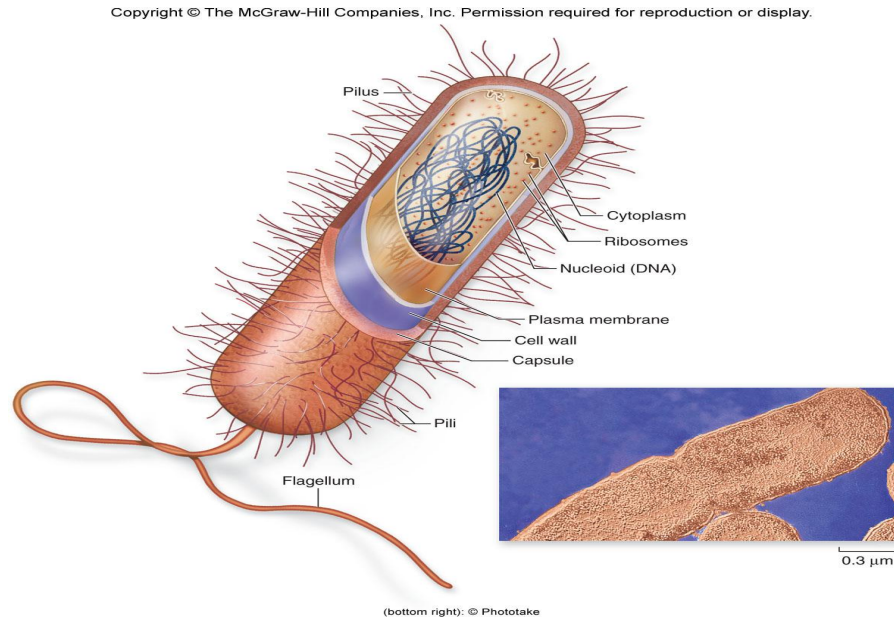


Red Blood Cell

Cell Types

Prokaryotic

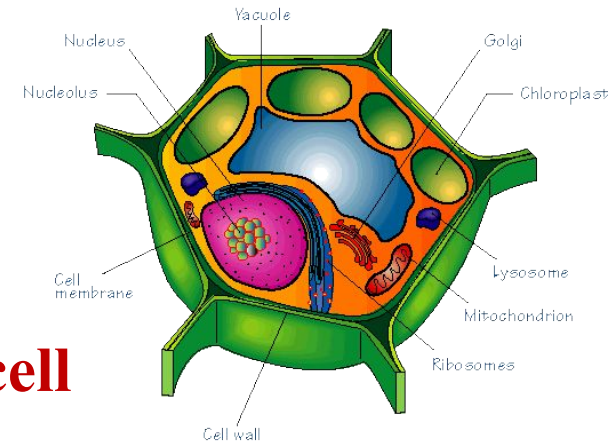
- Do not have structures surrounded by membranes
- Few internal structures
- Single-cell organisms, Bacteria
- Prokaryotic cells lack a membrane-bound nucleus - genetic material is present in the nucleoid
- Archaea and bacteria (types of prokaryotes).



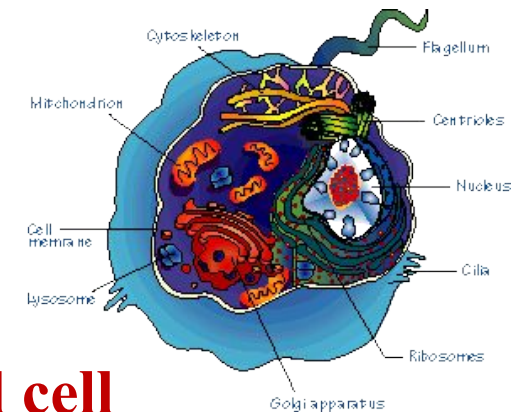
Eukaryotic

- Contain organelles
- Contain membrane bound nucleus, cytoskeleton for support and to maintain cellular structure.

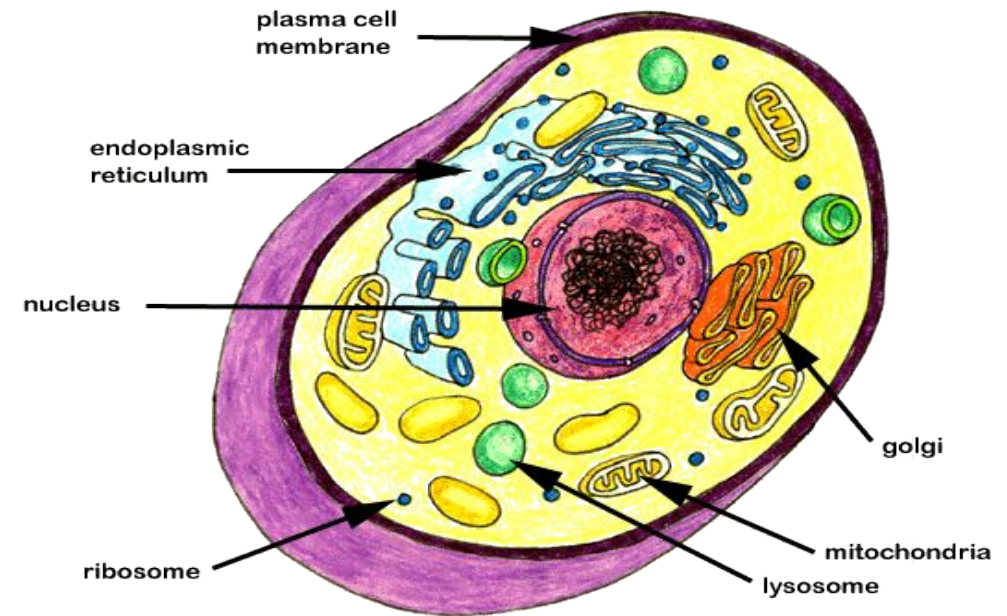
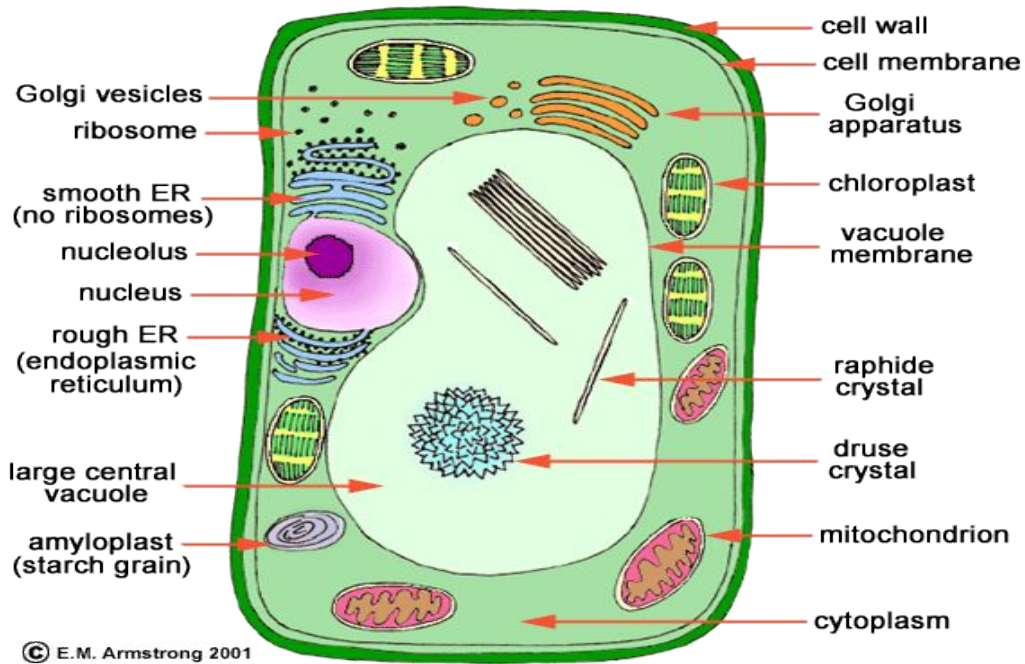
Plant cell

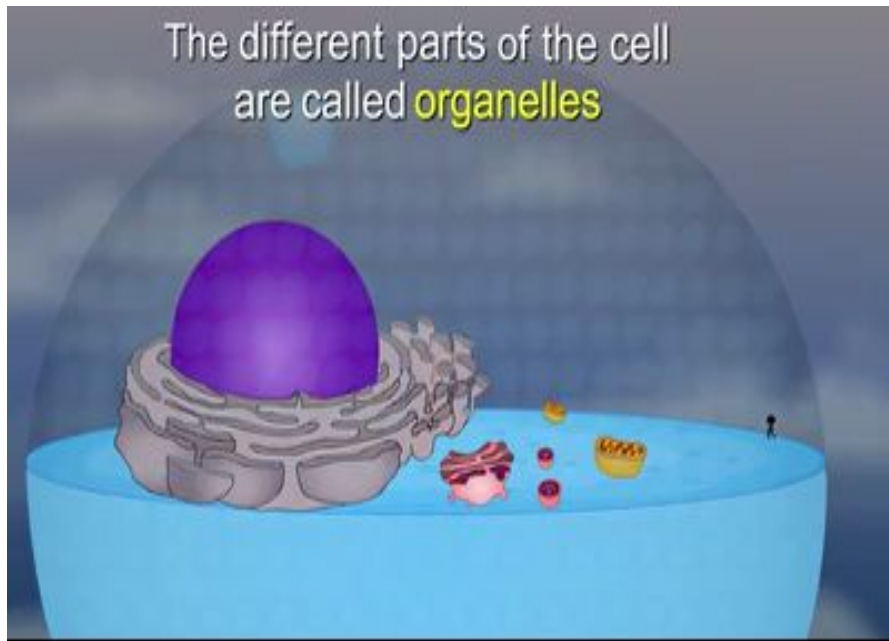


Animal cell

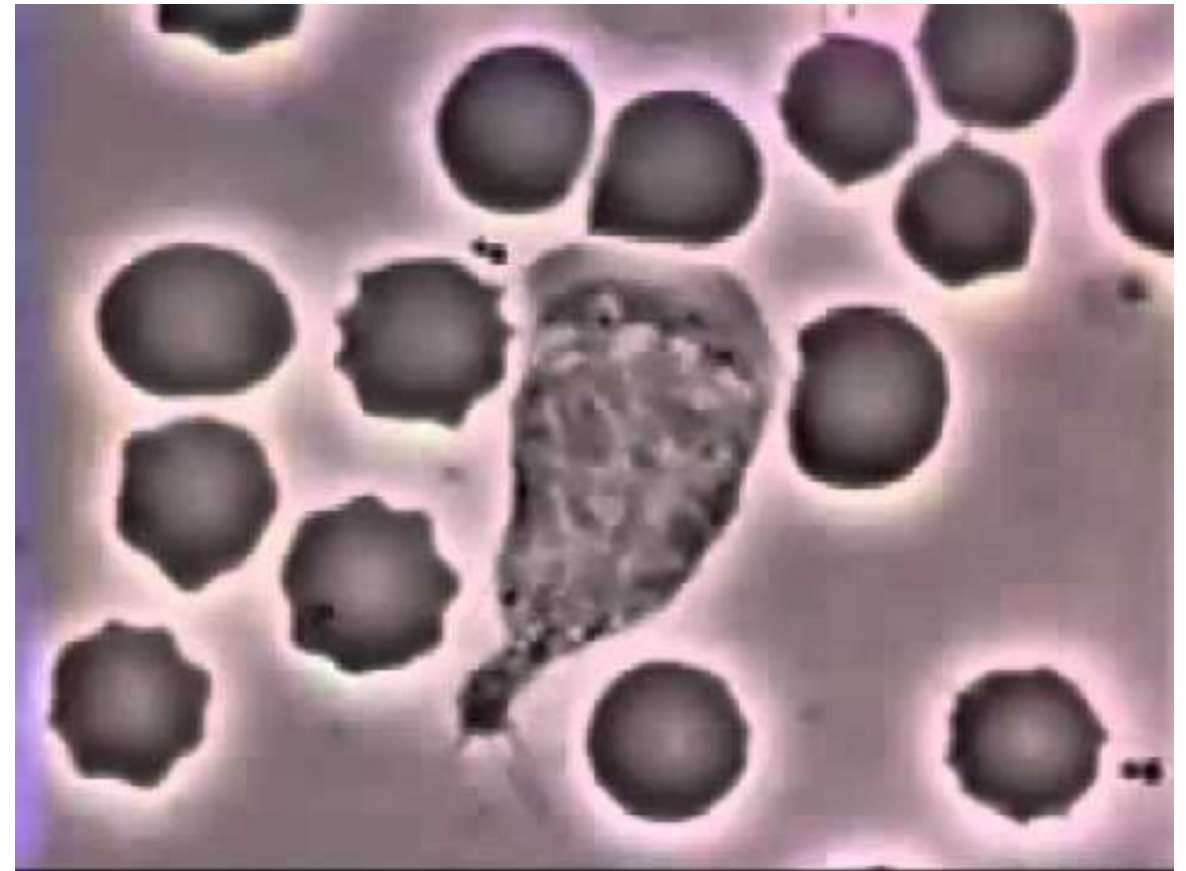


Plant and animal cell





Cellular organelles

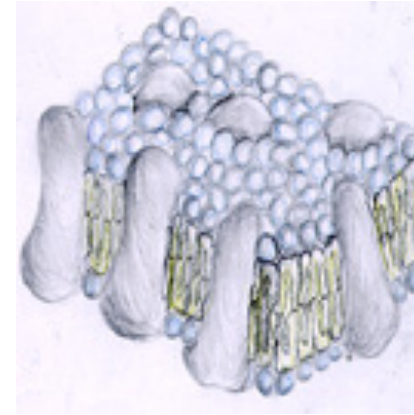


White blood cells (eukaryotes) chase bacteria (prokaryotes) with incredible agility despite their relative sizes.

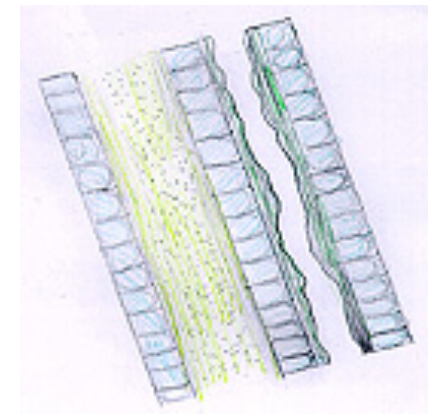
Functions of cell

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- ❑ **Cell membrane** – Outer membrane of cell that **controls movement** in- and out-of-the cell. It is made up of a **phospholipid bilayer**. It contains many types of membrane proteins.
- ❑ **Cell wall** – Mostly found in **plant cells** and bacteria. It supports and protects cells.
- ❑ **Endoplasmic reticulum** - **Moves materials around in cell**. **Smooth** type: **lacks ribosomes**; **Rough** type (pictured): **ribosomes embedded in surface**.



Cell membrane



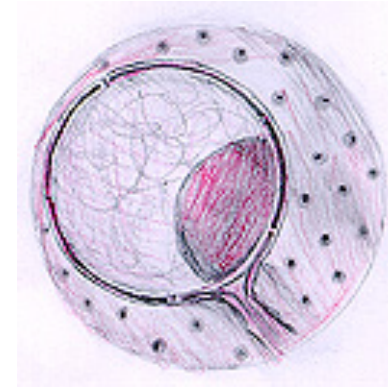
Cell wall



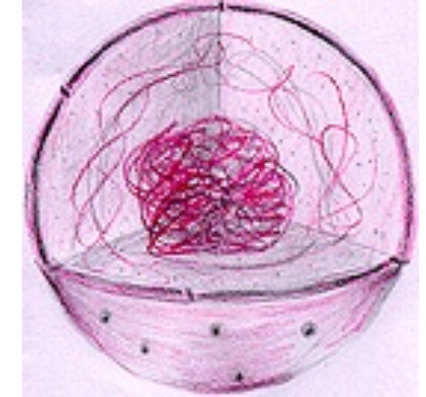
**Endoplasmic
reticulum**

Functions of cell

- ❑ **Nucleus** - stores the genetic material of the cell in the form of multiple, linear chromosomes. It is surrounded by a nuclear envelope composed of 2 phospholipid bilayers. In chromosomes, DNA is organized with proteins to form chromatin.
- ❑ **Nuclear membrane** – Surrounds nucleus; made of two layers; openings allow material to enter and leave nucleus.
- ❑ **Nucleolus** – Inside nucleus; contains RNA to build proteins.
- ❑ **Ribosomes** - the site of protein synthesis in the cell. It is composed of ribosomal RNA and proteins. It is found within the cytosol of the cytoplasm and attached to internal membranes.



Nuclear membrane



Nucleolus



Ribosomes

Functions of cell

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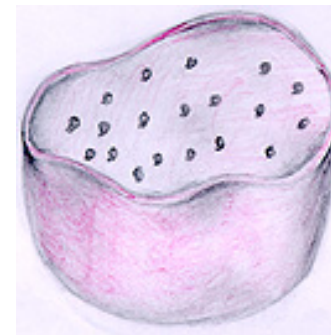
- ❑ **Mitochondria** – Powerhouse of the cell; produces energy through chemical reactions – breaking down fats & carbohydrates; Controls level of water and other materials in cell; Recycles and decomposes proteins, fats, and carbohydrates. It is filamentous or globular in shape.
- ❑ **Golgi bodies** – Protein 'packaging plant'; Move materials within the cell; Move materials out of the cell.
- ❑ **Lysosome** – Digestive 'plant' for proteins, fats, and carbohydrates; Transports undigested material to cell membrane for removal; Cell breaks down if lysosome structure is disrupted.
- ❑ **Vacuoles** - Membrane-bound sacs for storage, digestion, and waste removal; Contains water solution; Help plants maintain shape.



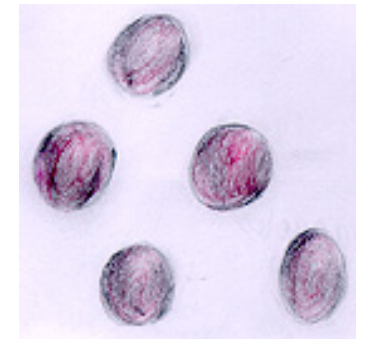
Mitochondria



Golgi bodies



Lysosome



Vacuoles

Summary

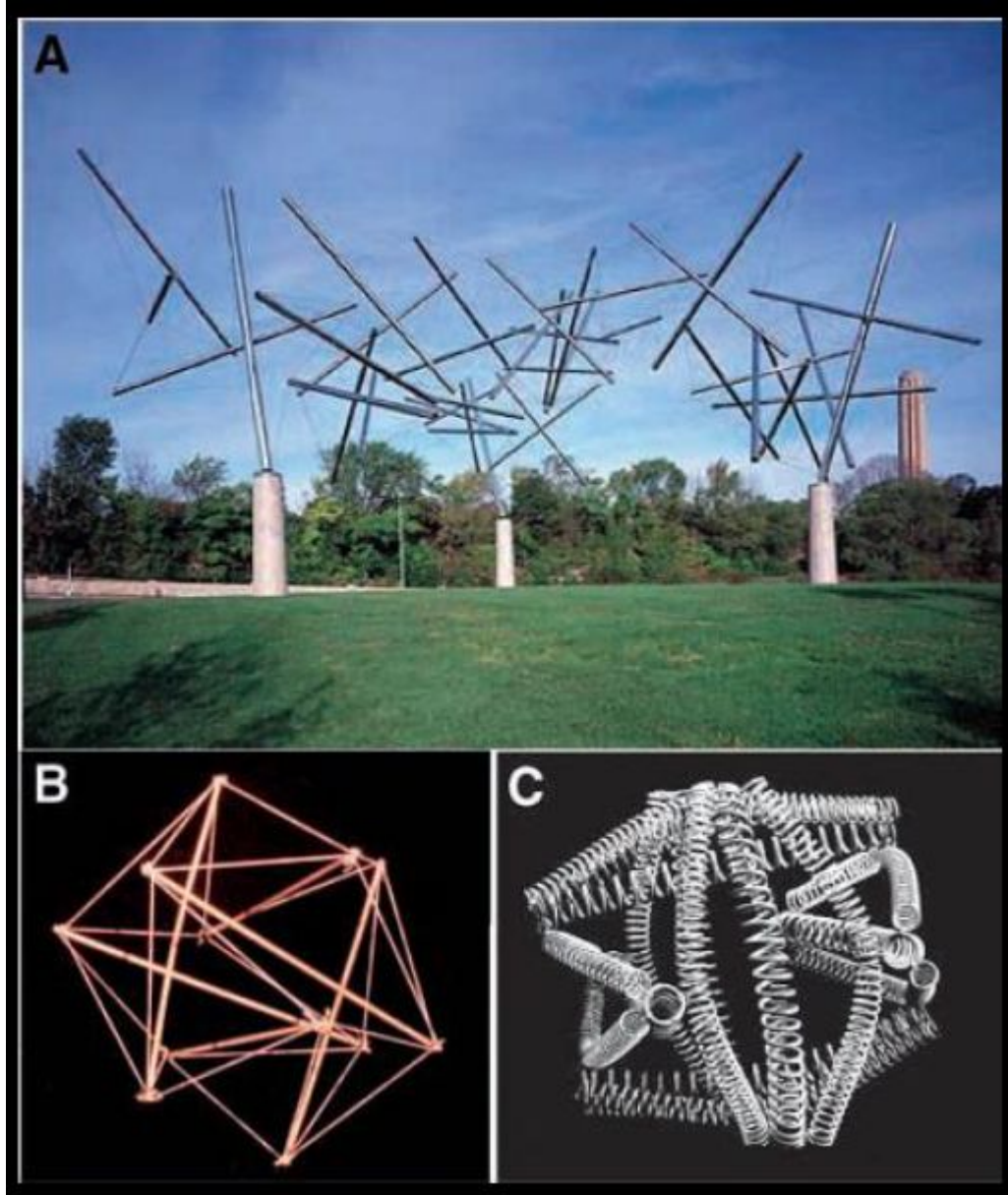
Components

- Plasma membrane
- Cytosol
- Mitochondria
- Endoplasmic reticulum
- Golgi apparatus
- Lysosomes
- Cytoskeleton
- Nucleus

Major functions

- Transport of ions and molecules.
- Metabolism of carbohydrates, lipids and amino acids
- Energy production
- Synthesis of proteins and lipids
- Modification and sorting of proteins
- Cellular digestion
- Cell morphology and cell motility
- DNA synthesis and repair

Tensegrity



Tensegrity: building principle, first described by R. Buckminster Fuller (architect) structures that stabilize their shape by continuous tension or ‘tensional integrity’ rather than by continuous compression.

Tensegrity structures:

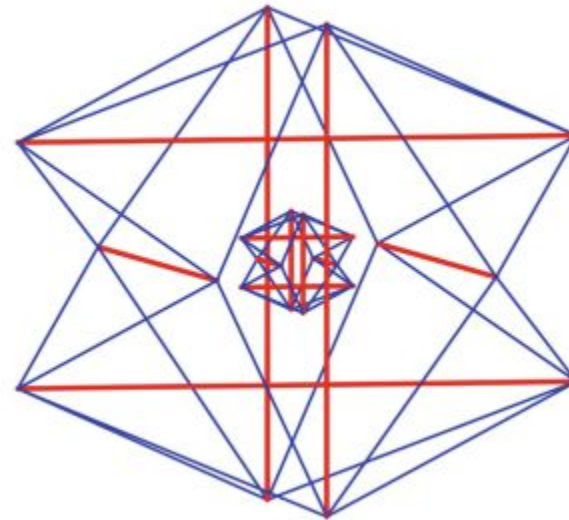
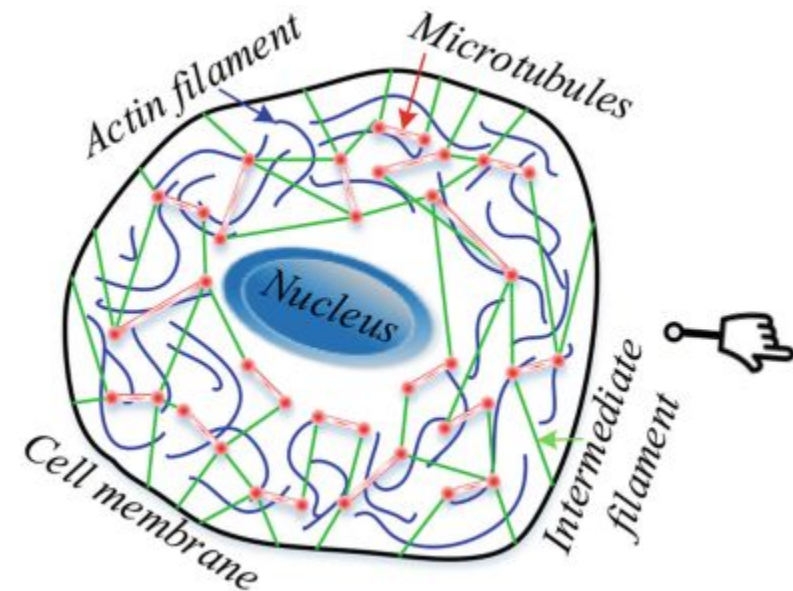
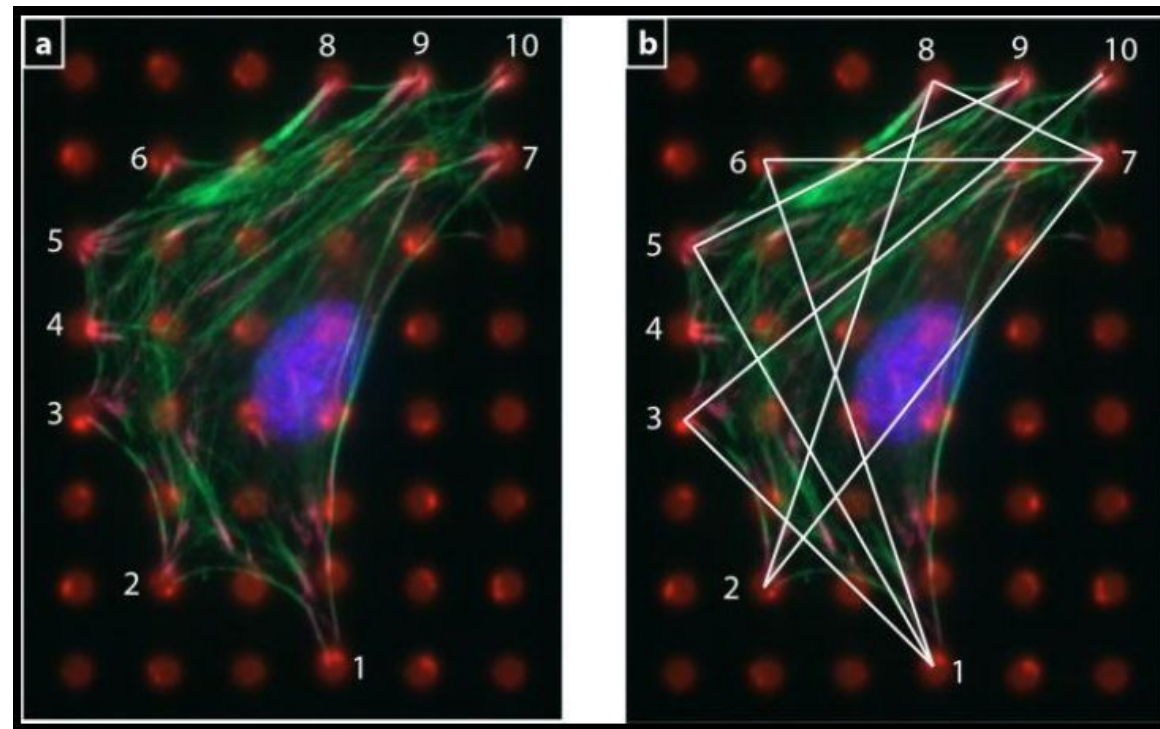
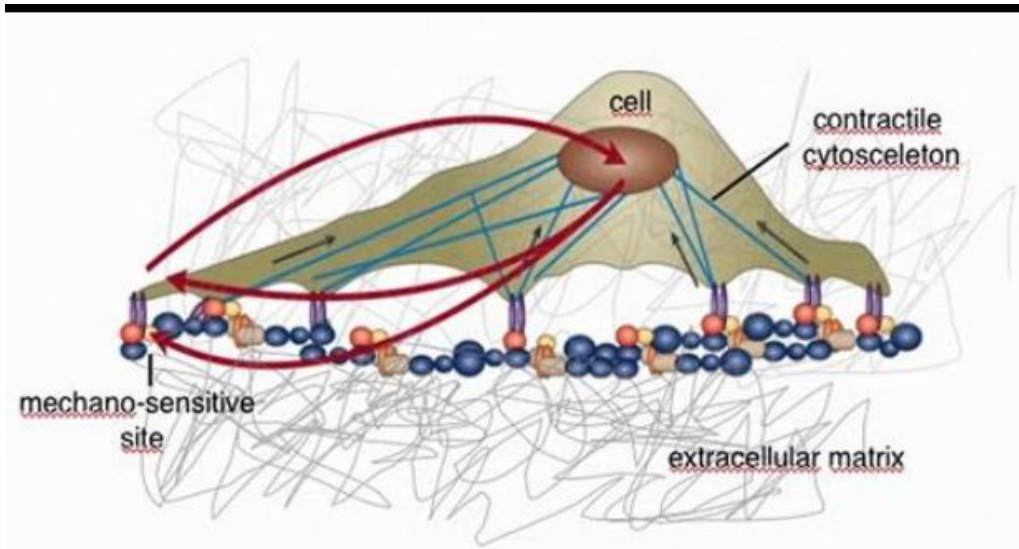
(A) Triple crown, a tensegrity sculpture, that is composed of stainless-steel bars and tension cables.

(B) A tensegrity sphere composed of six wood struts and 24 white elastic strings.

(C) The same tensegrity configuration as in B constructed entirely from springs with different elasticity.

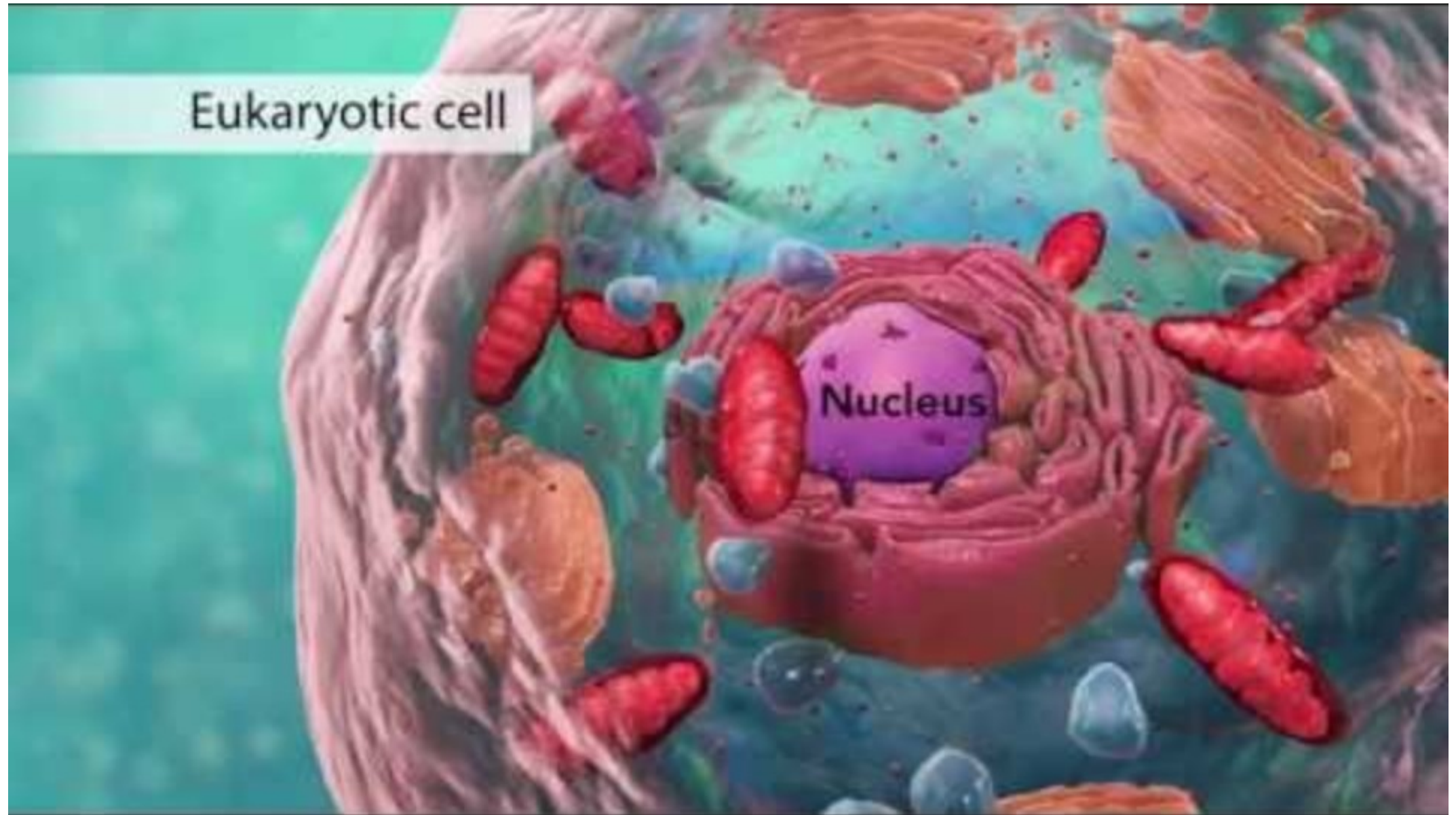
Cellular Tensegrity Model

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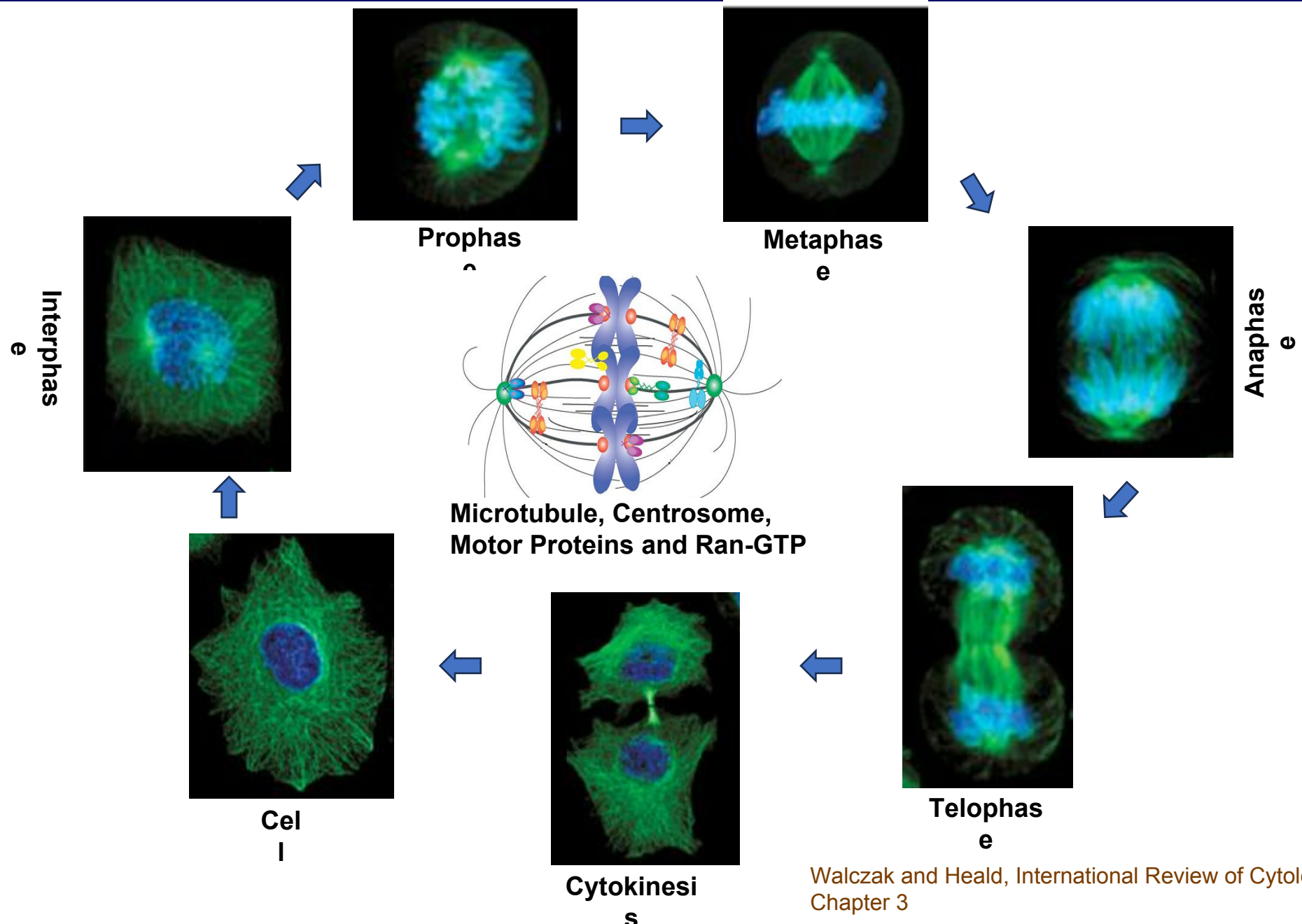


- Tensional forces are borne by cytoskeletal microfilaments and intermediate filaments.
- Balanced by interconnected structural elements that resist compression

Summary

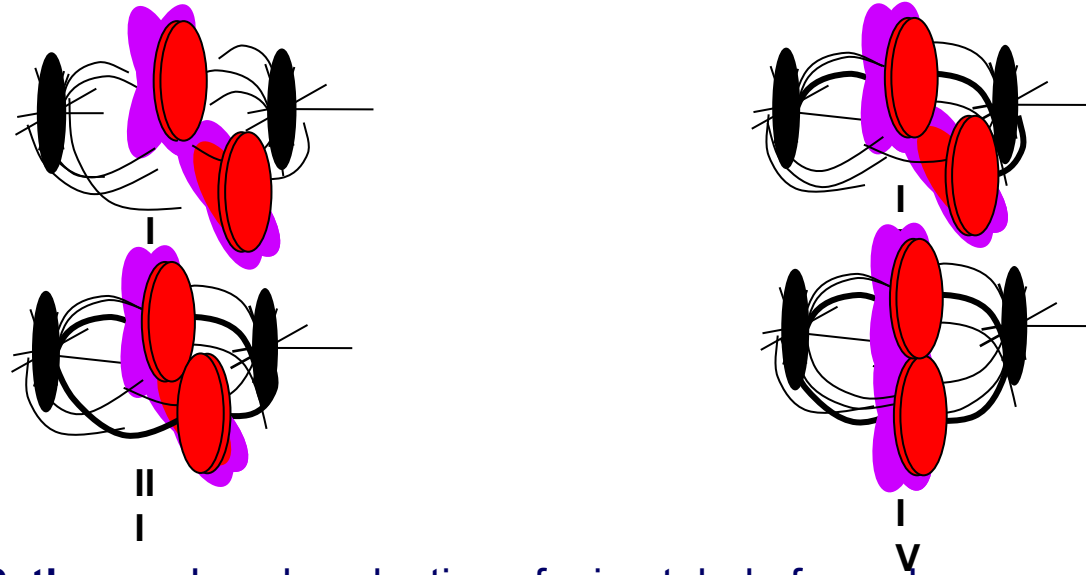


Bioengineering in : Mitotic Cell Division

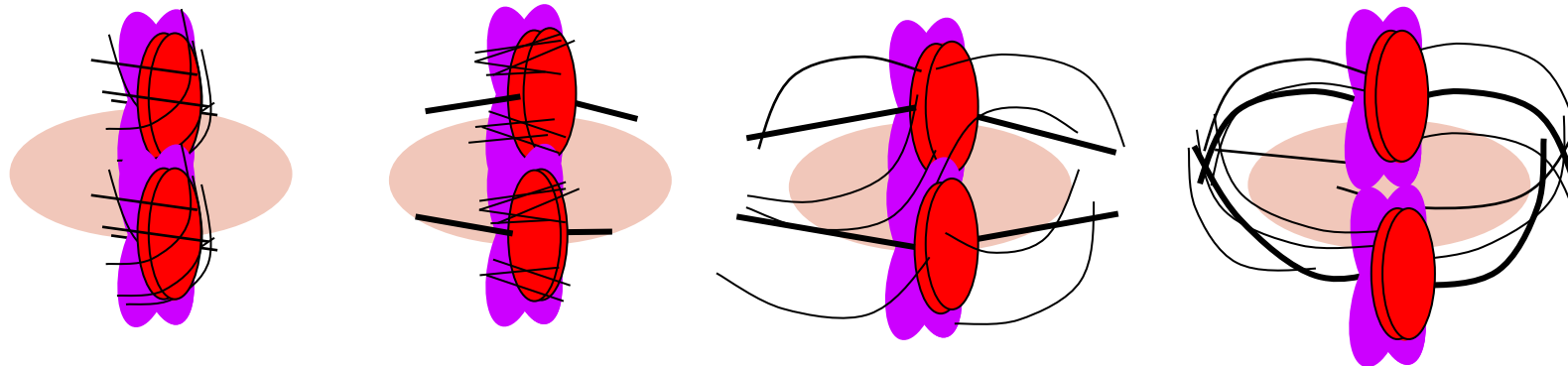


Chromosome Segregation Pathways

Search and Capture pathway: Microtubule generation from Centrosome dynamic Instability



Chromatin Pathway: Local nucleation of microtubule from chromosome
Ran-GTP helps in nucleation
Role of different types of motor proteins

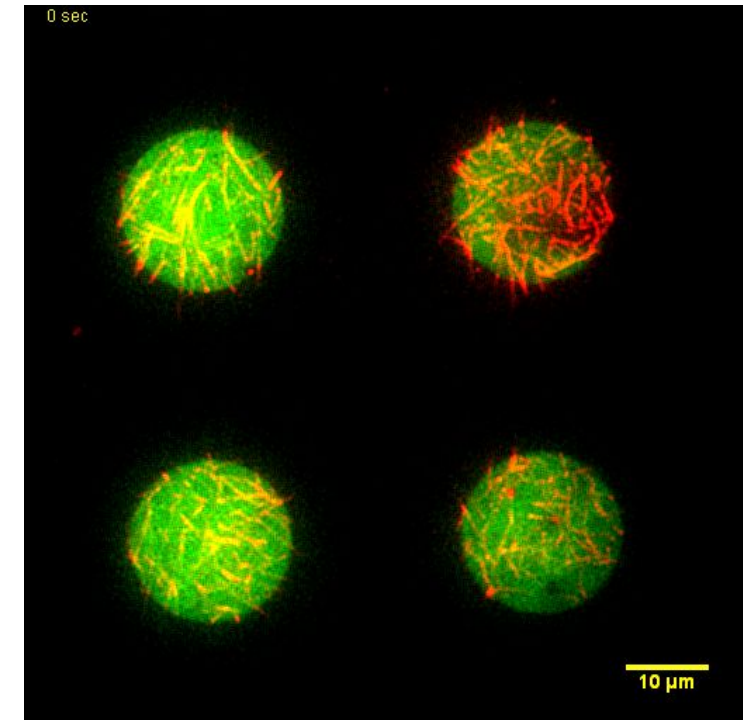
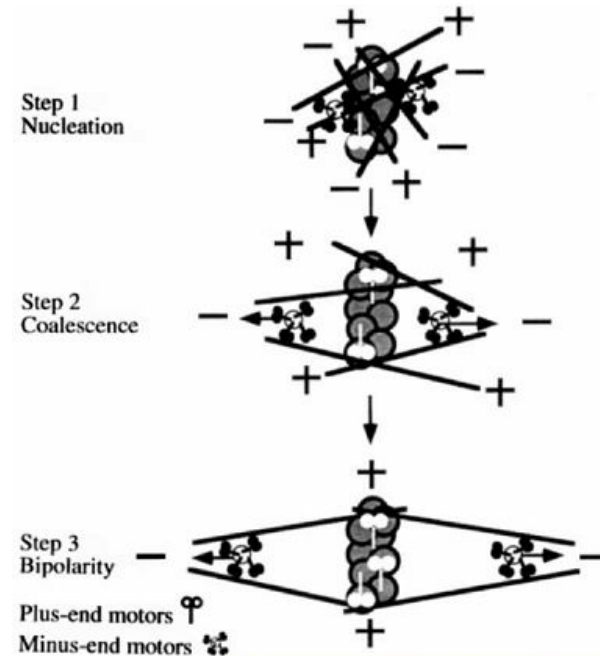
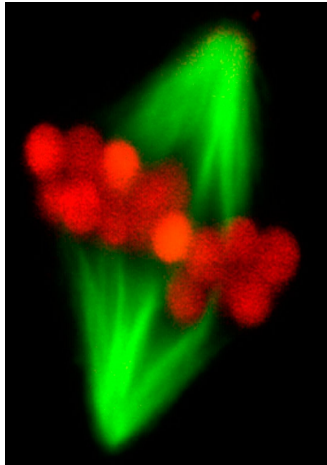


Motivation- Microtubule on chip

- Nucleation of microtubule around the DNA coated bead in absence of centrosome and kinetochore

Heald *et al.*, *Nature* 1996

Dinarina *et al.*, *Cell*, 2009



- Microtubule nucleation and mesoscale organizations on 2D micropattern surface

Ghosh *et al.*, *ACS Chem. Biol.* 2013

Open Question

- **How to reconstitute chromosome and centrosome by biomimetic approach for measuring the force during spindle formation?**
- **How to quantify or measure the mitotic force on chromosome during cell division?**

Objectives

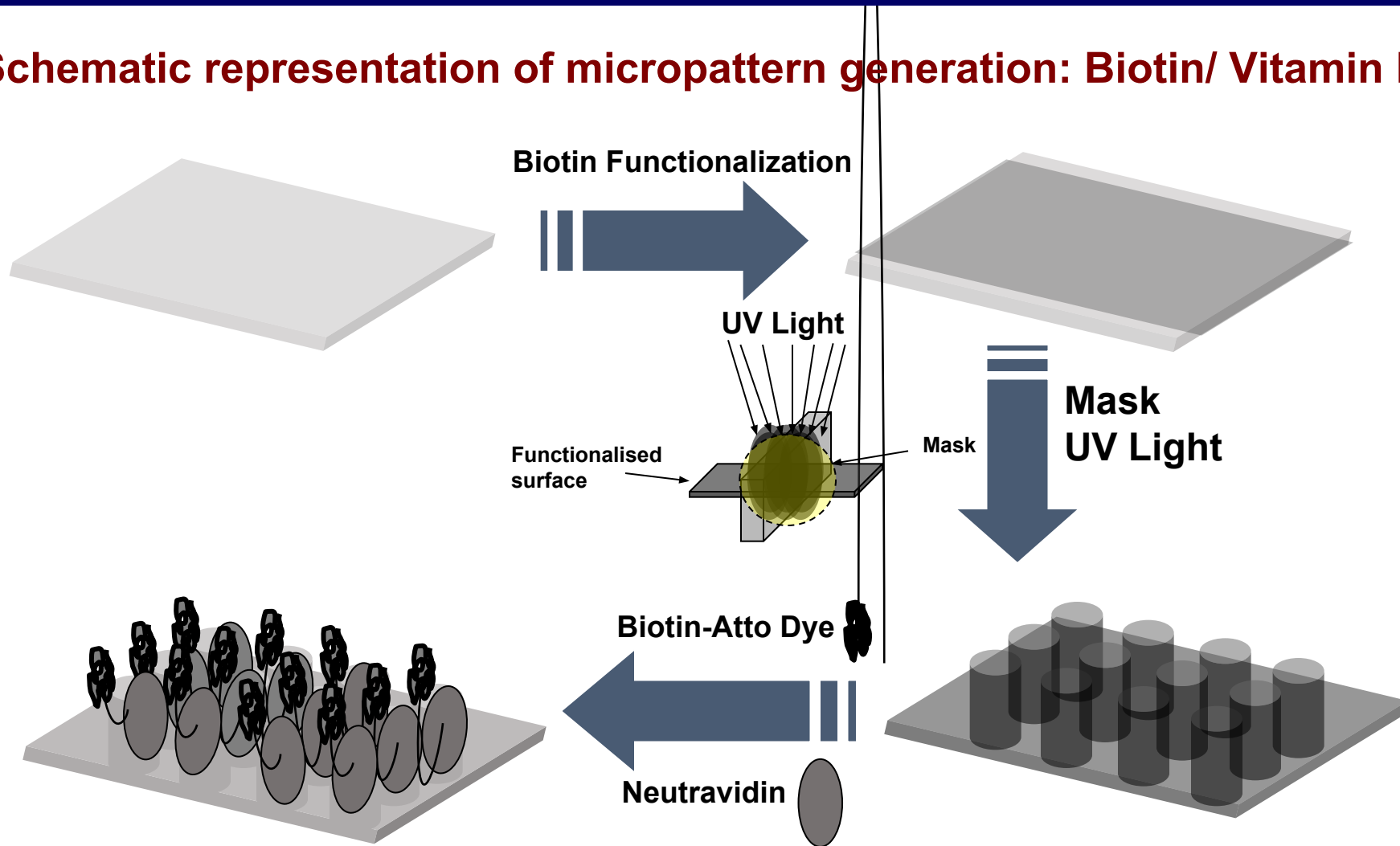
- **Reconstitution of chromosomal surface by generation of novel biocompatible 2D micropattern surface**
 - Mimicking chromosomal surface by immobilization of GMP-CPP short microtubules or biotinylated DNA
- **Reconstitution of chromosomal and centrosomal surface by generation of novel biocompatible 2D micropattern surface simultaneously**
- **Measurement of nucleation force on both biomimetic chromosome and centrosome surface by nucleating microtubules from those areas by AFM**

Methodology

- Preparation of biotin, dual (biotin & Tris-NTA) functionalized Surface
- Generation of biotin and dual (Biotin & Tris-NTA) micropattern surface and imaging
- Biotin labelled and unlabelled Alexa 568 GMP-CPP Microtubule preparation
- Selective immobilization of biotinylated short microtubules and biotinylated DNA on biotin micropattern
- Selective immobilization XMAP215-His Tris-NTA micropattern among dual pattern
- Immobilization of BCCP-Kin401 (biotin tagged kinesin401) and organization of Alexa 568 GMP-CPP stabilized microtubules
- Measurement of force generated by nucleated microtubules from micropattern surfaces by Atomic Force Microscopy
- Data Analysis

Preliminary Studies and Initial Results

Schematic representation of micropattern generation: Biotin/ Vitamin H



Open Question

- How biomolecules/proteins work as a system to organize complex structure in biology (e.g. microtubules around the chromosomes)

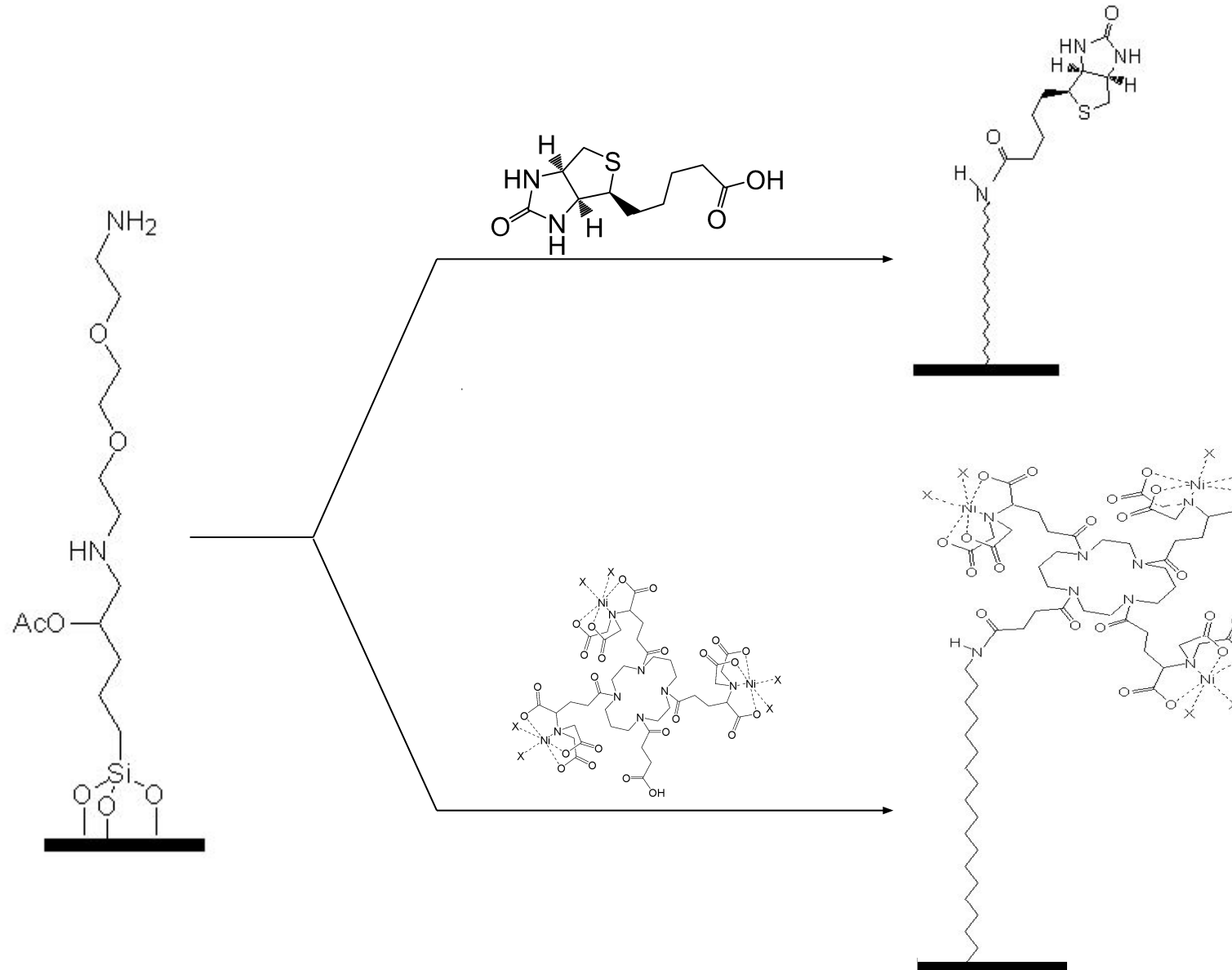
Our Approach

□ *Generation of novel bioinspired micropattern surface*

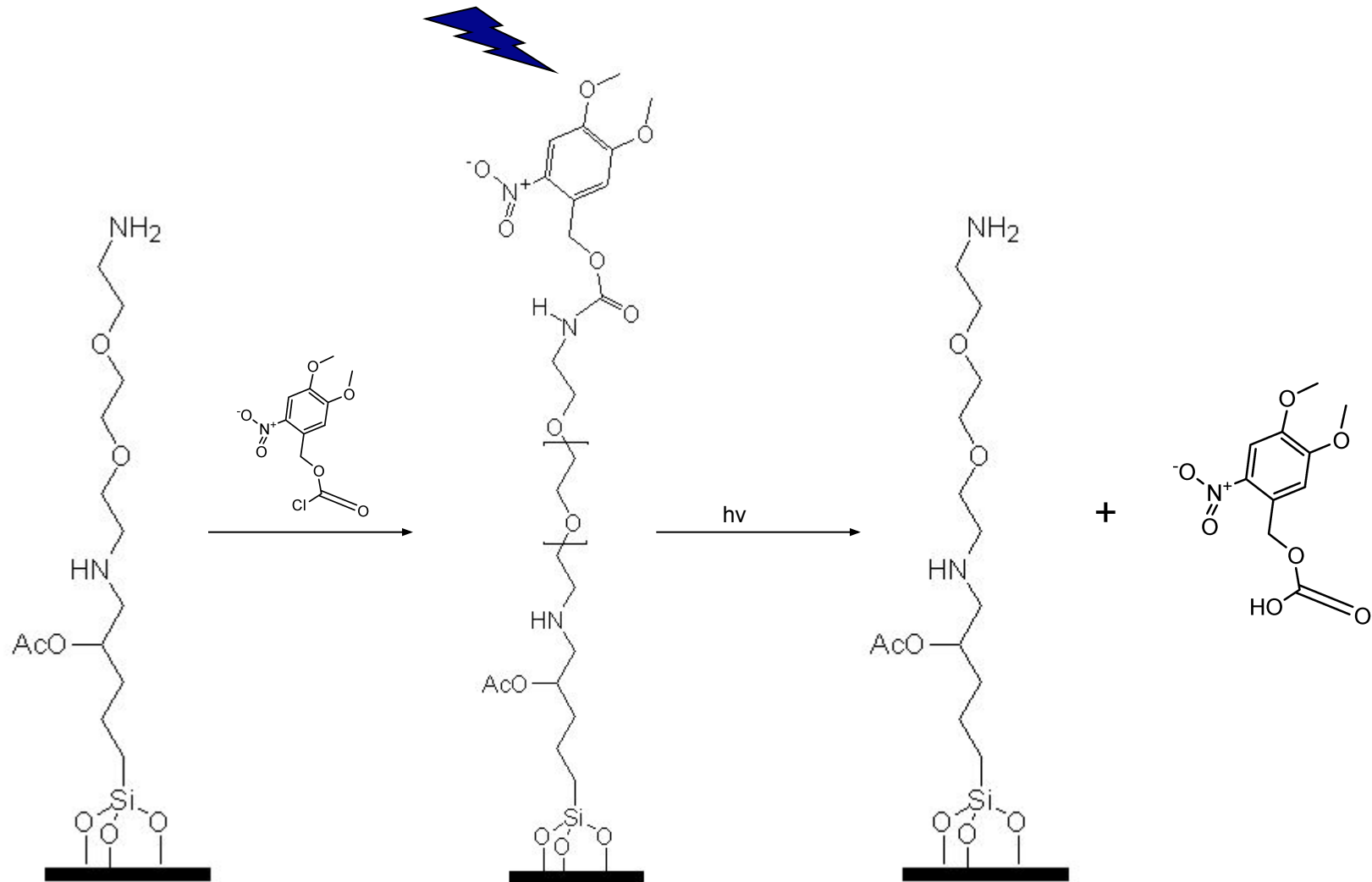
- Organization of biomolecules on micropattern surface
- Introduction of 2D Model



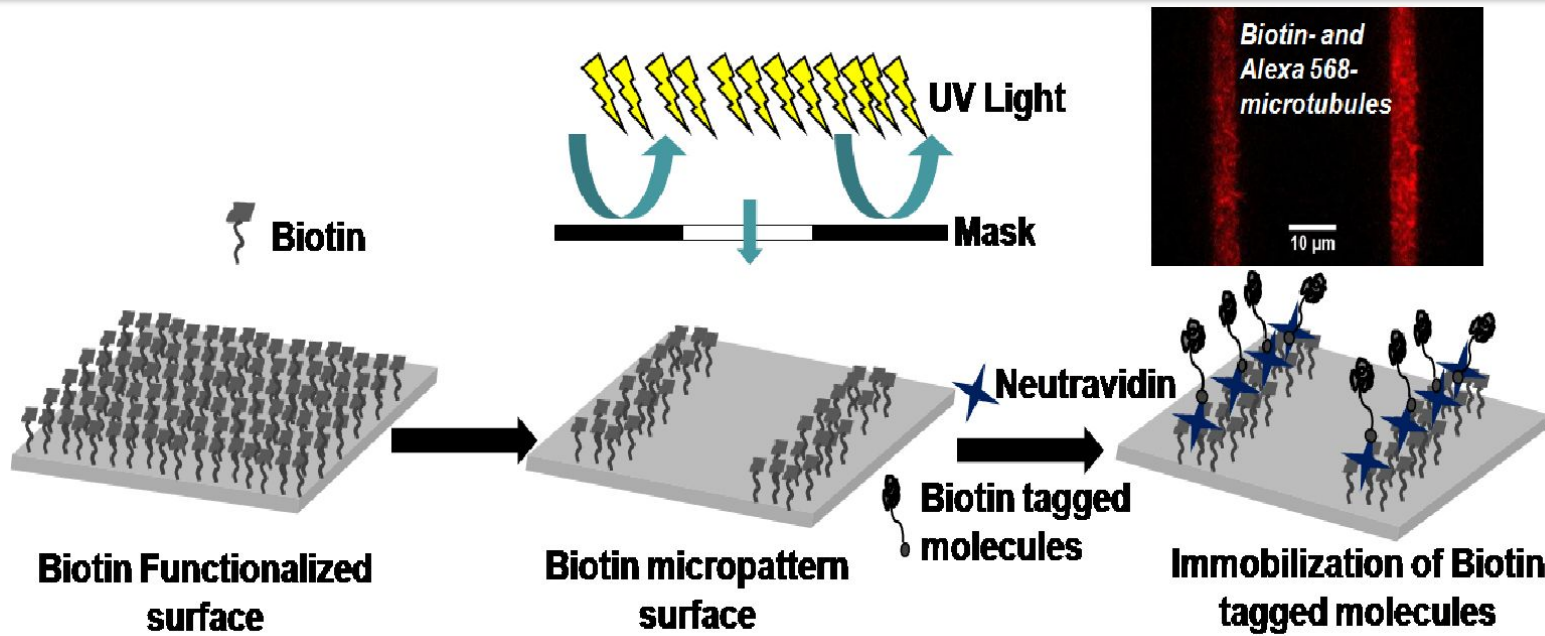
Biotin and Tris-NTA molecule for protein capturing



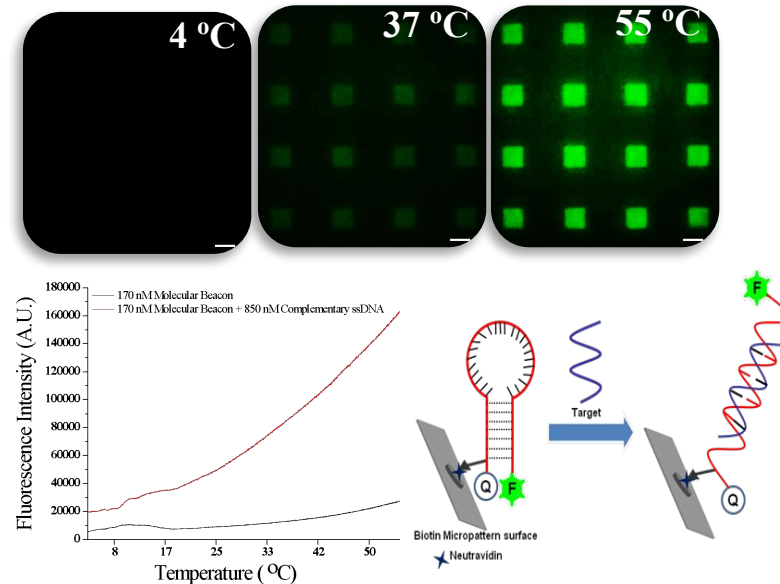
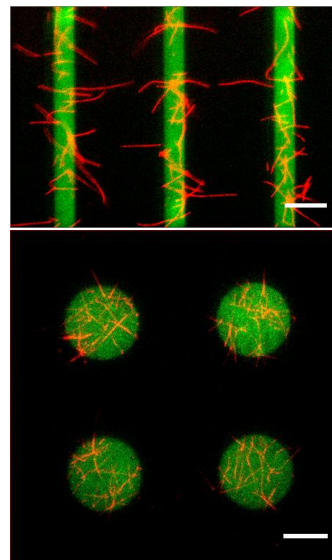
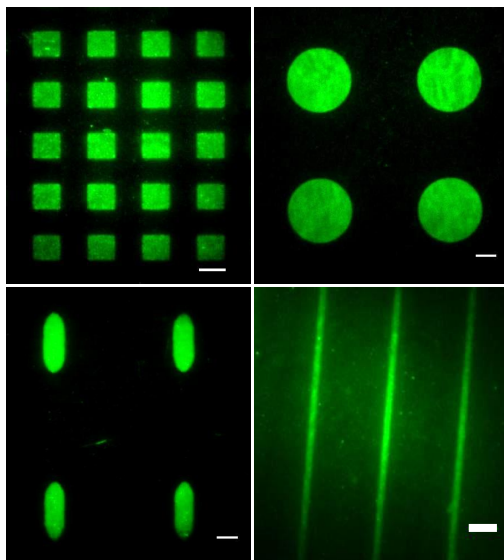
NVOC Chemistry



Schematic View of Biotin Micropattern Generation

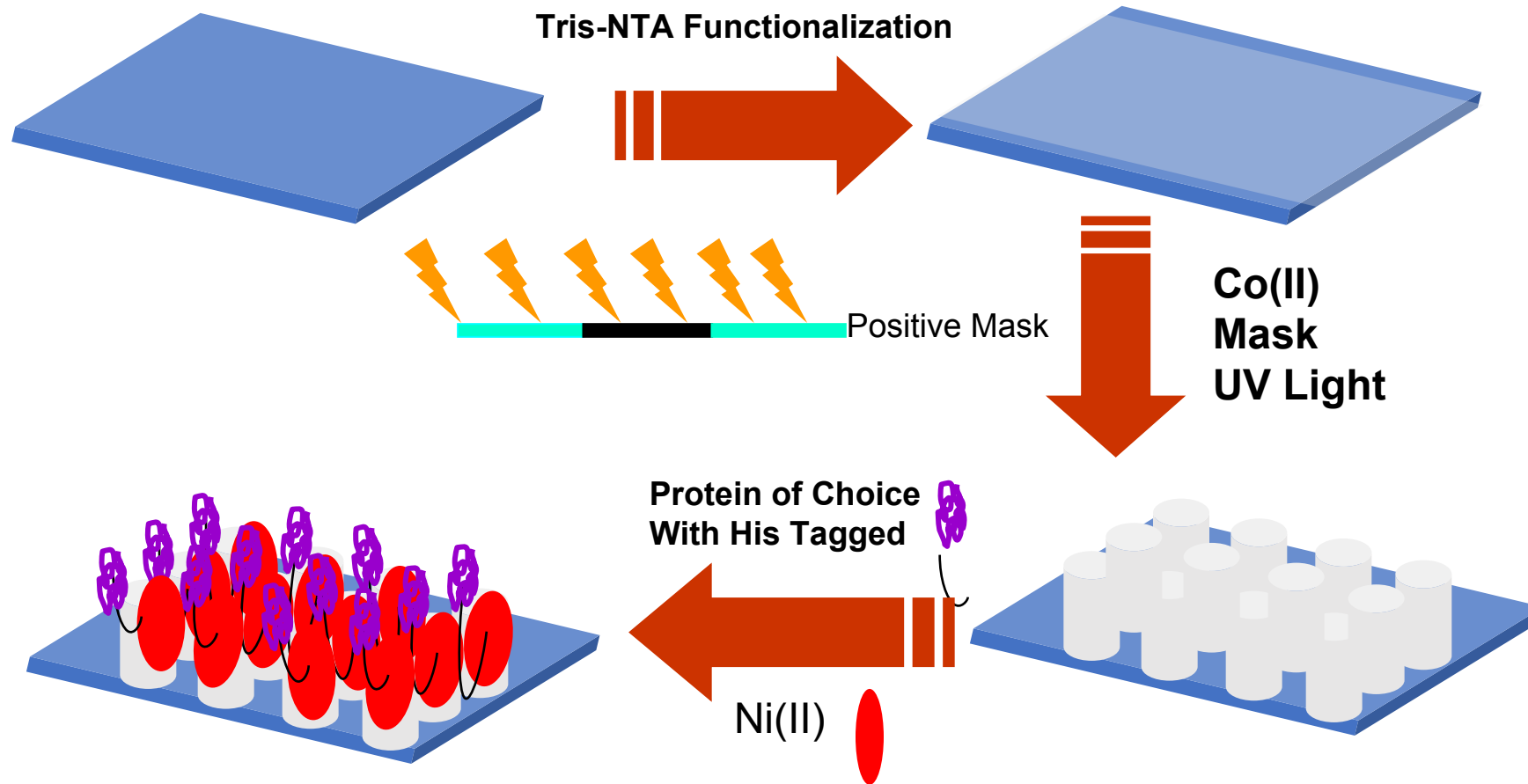


Results & Applications



Generation of Tris-NTA micro-pattern surface

Schematic Overview of the Surface Patterning and Protein Immobilization

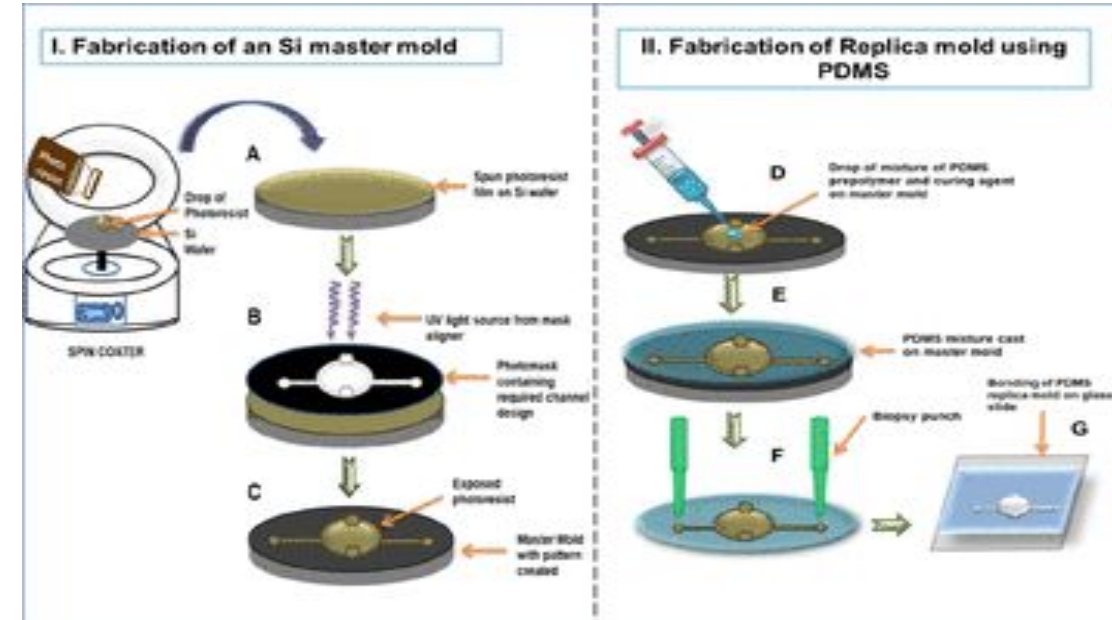
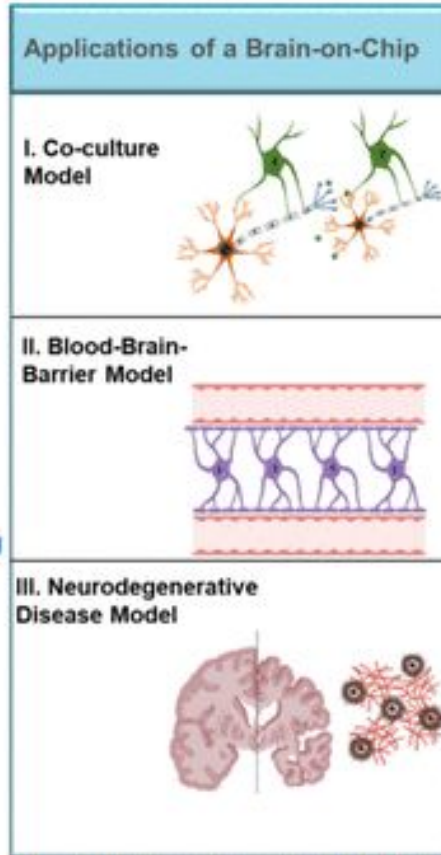
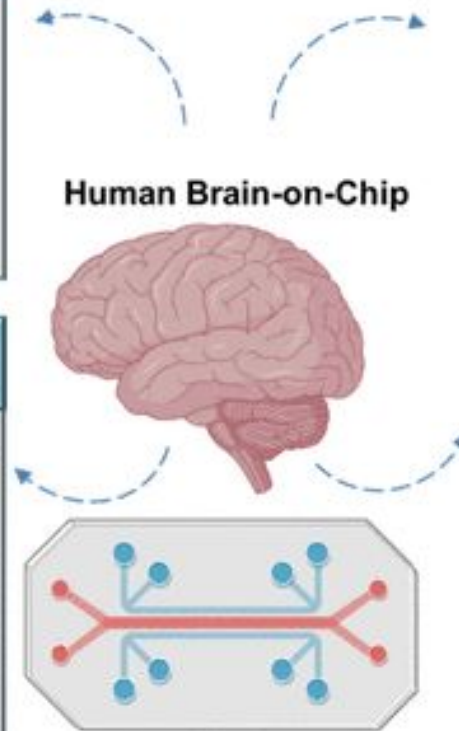
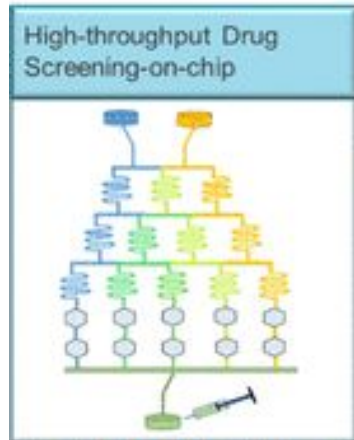
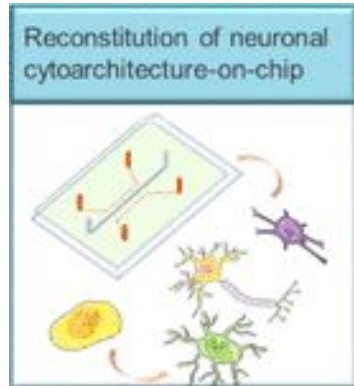


Can we make both Tris-NTA and biotin functionality on same micropattern and individually on side-by-side micropattern?



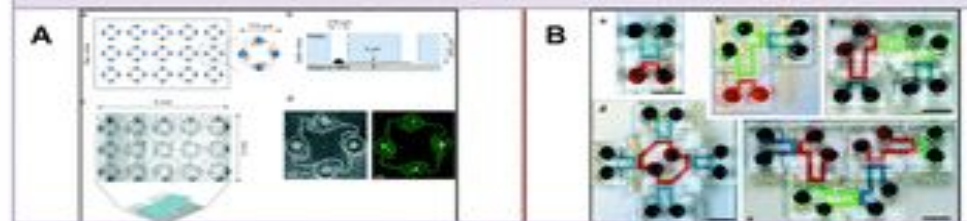
Why do we need those micropattern surface?

BRAIN-ON-CHIP



III. Few Applications of the Assembled Microfluidic Device

(i) Reconstruction of neuronal networks in microfluidic device



(ii) Brain stimulation-on-chip



- Traditional *in vitro* macroscopic cell culture techniques offer simple and straightforward methods.
- However, they often fall short of providing insights into the complex phenomena of neuronal network formation and the relevant microenvironments.
- To circumvent the drawbacks of conventional cell culture methods, recent advancements in the development of microfluidic device-based microplatforms have emerged as promising alternatives.
- Microfluidic devices enable precise spatiotemporal control over compartmentalized cell cultures. This feature facilitates researchers in reconstituting the intricacies of the neuronal cytoarchitecture within a regulated environment.

"Engineers are not going to replace biologists, nor should we try to. ... The value of having engineers in medicine is that we're trained in a different way and approach problems in a different way." --Marc Levenston

Thank you